

Kernel Methods

Dr. Krishna Bathula

Learning Objectives

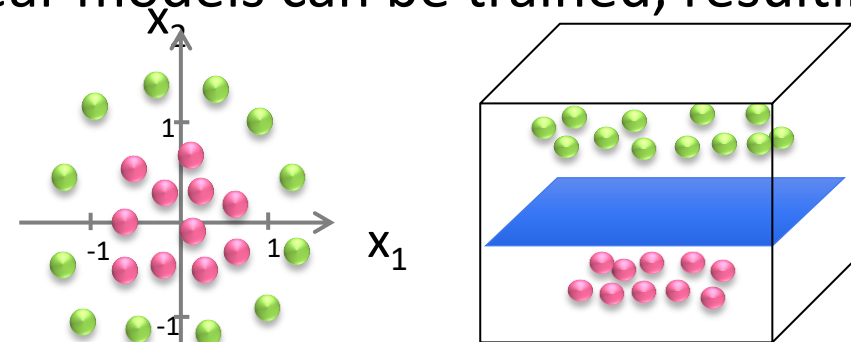
1. Introduction to Kernel methods
2. Types of Kernel functions
3. Kernel Function in Support Vector Machines (SVMs)
4. Advantages and disadvantages of Kernel methods
5. Applications of Kernel methods
6. Conclusion

Kernels Methods - Introduction

Kernel methods are a class of **algorithms** used in machine learning **for nonlinear** regression, classification, and clustering.

They are based on **transforming data into a higher-dimensional** feature space using a kernel function.

Kernel methods do this by mapping the input space of the data to a higher dimensional feature space, in which simple linear models can be trained, resulting in efficient, low bias and low variance models.



In this feature space, it becomes easier to separate or classify data that is not linearly separable in the original space.

Kernels Function

In kernel methods, the data is represented as a **set of vectors** in a feature space.

The kernel function is a **similarity measure between two vectors** in this space.

Kernel methods use kernels (or **basis functions**) to map the input data into a different space.

They **aim to find a linear function** in the feature space that separates or classifies the data.

Kernel methods are most suited for **small to medium datasets** and in problems where the ability to explain the results is important.

Kernels Density Estimation

Kernel density estimation (KDE) is a non-parametric method to estimate a **random variable's probability density function**.

The method estimates the density of a distribution by smoothing a set of data points with a kernel function, which is a **weighted function** centered on each data point.

The basic idea of KDE is to estimate the density of a distribution by adding up the contributions of a set of kernels, where each kernel is a smoothed version of the distribution centered at a particular point.

The **shape of the kernel** function determines the **degree of smoothing**, and the bandwidth of the kernel controls the width of the smoothing.

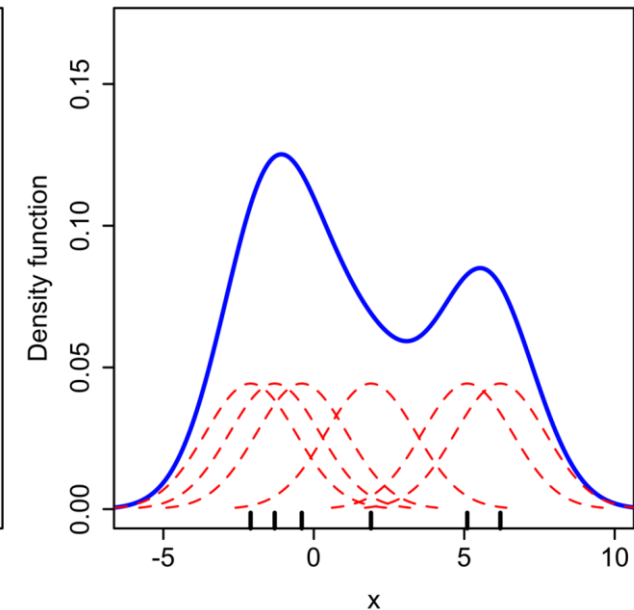
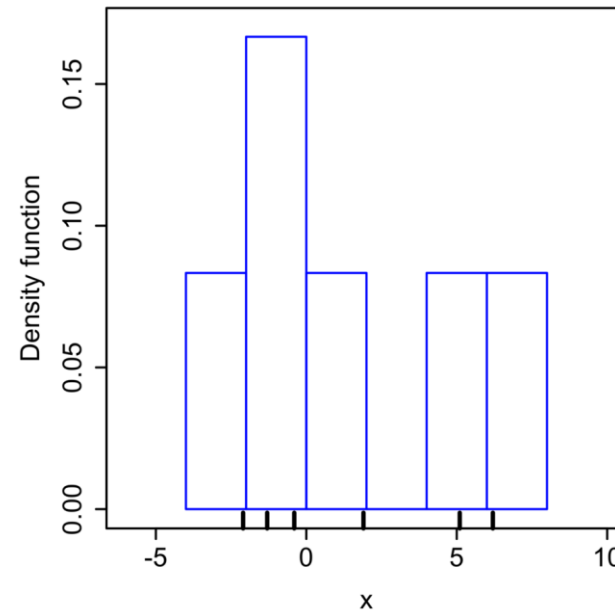
Kernels Density Estimation

Apply a Kernel on the data points $\mathbf{x}_i$ and add all the kernels. The result is the distribution where these $\mathbf{x}_i$ are close together.

$$\hat{p}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

Datapoints:

(-1.10, -0.75, -0.20, 1, 5, 5.1)



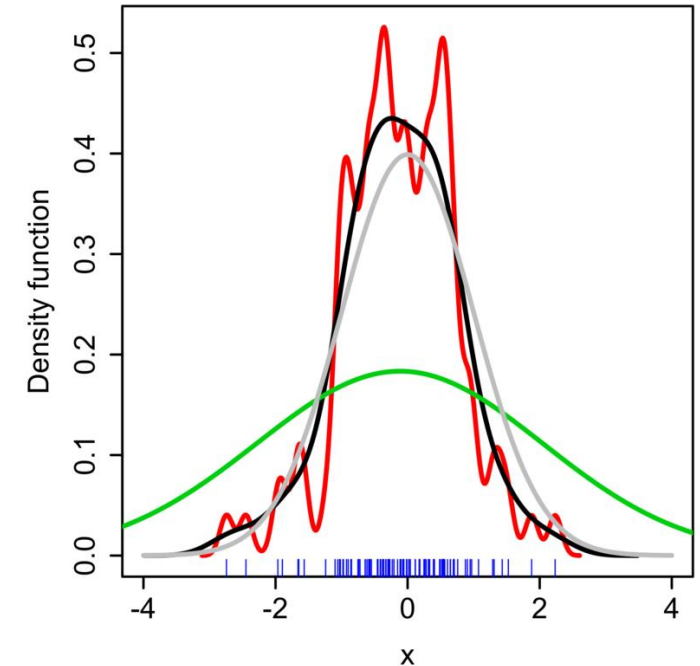
Kernel Smoother - KDE

A **Kernel Smoother** is a statistical technique to estimate a real valued function as the **weighted average** of neighboring observed data.

The **weight** is defined by the kernel, such that closer points are given higher weights.

KDE is widely used in data analysis, pattern recognition, and machine learning, where it is often used for density estimation, clustering, and classification.

KDE is a powerful and flexible technique that can handle a wide range of data types, and it does not require any assumptions about the underlying distribution of the data.



Kernel Tricks

The kernel trick works by finding a function (kernel) that calculates the similarity between pairs of data points in the higher-dimensional space.

This similarity score can be used as a measure of how close or how far apart the points are, and it can be used to construct decision boundaries or regression curves.

One common example of a kernel function is the radial basis function (RBF) kernel, which measures the distance between two data points and assigns a weight based on this distance.

Other examples include the polynomial kernel and the sigmoid kernel.

Kernel Tricks

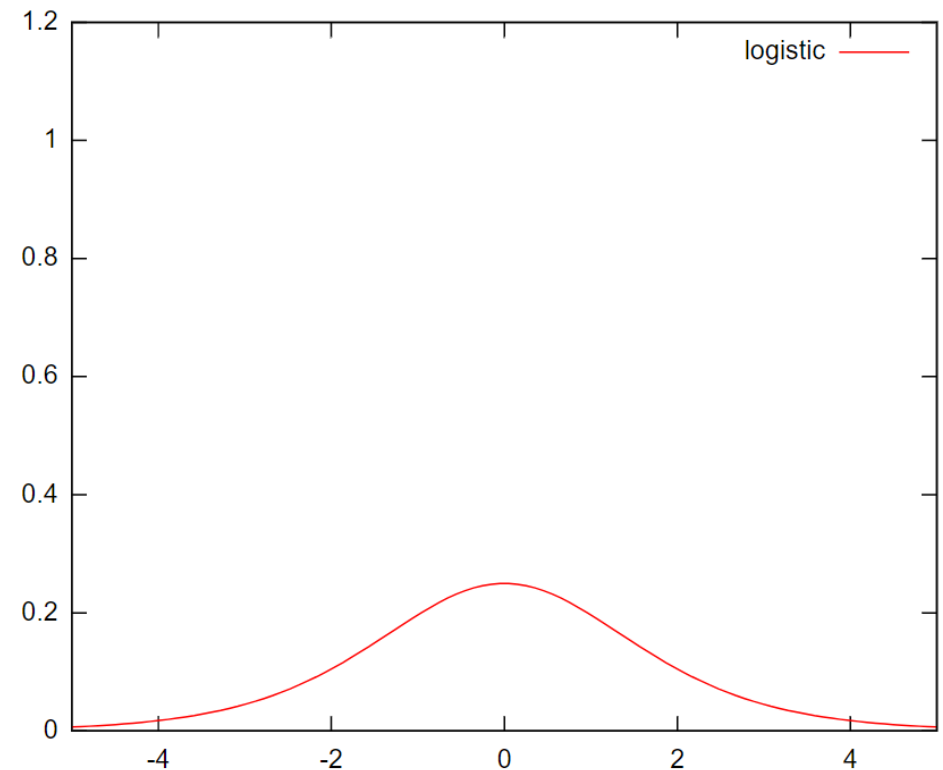
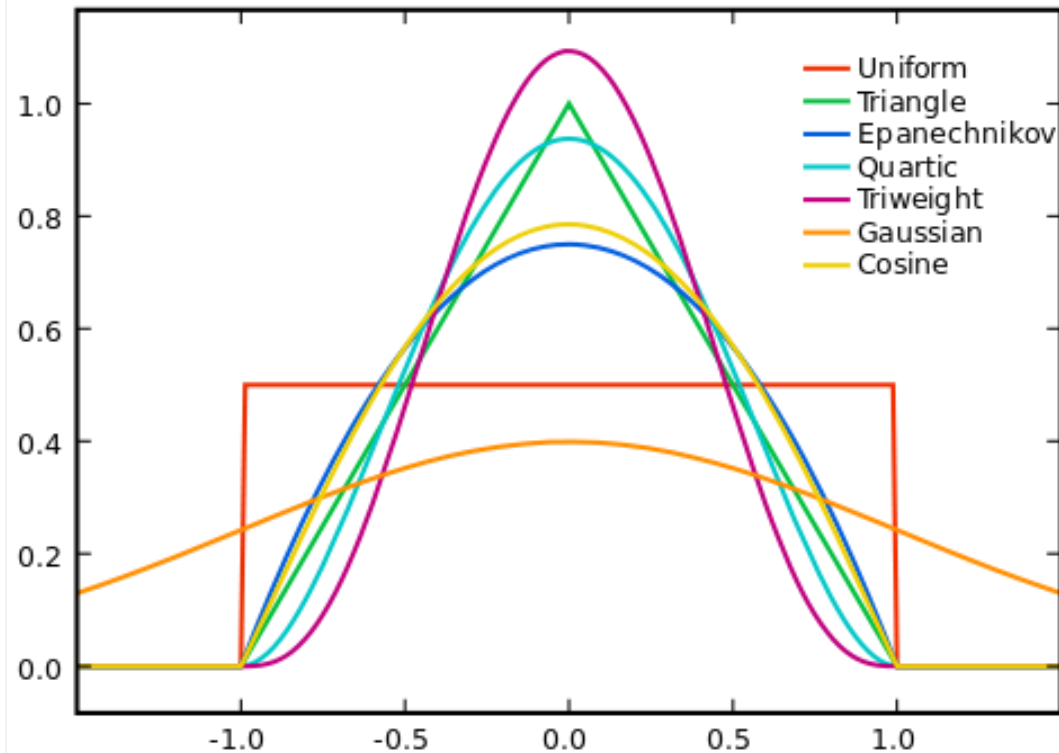
The **advantage of kernel tricks** is that they can allow algorithms like **support vector machines (SVMs)** to handle **non-linearly separable data** without the need for **explicit feature mapping**.

This can greatly simplify the problem and **reduce computational costs**.(ex:)

Additionally, kernel tricks can be used in many other machine learning algorithms such as **kernel principal component analysis (PCA)** and **kernel ridge regression**.

Kernel Functions

Several types of kernel functions are commonly used: uniform, triangle, Gaussian, quadratic and cosine.



Kernels Function Types & Expressions

- **Linear kernel: $K(x, y) = x^T y$**
- **Polynomial kernel: $K(x, y) = (x^T y + c)^d$**
- **Gaussian kernel: $K(x, y) = \exp(-||x - y||^2 / (2 * \sigma^2))$**
- **Laplacian kernel: $K(x, y) = \exp(-||x - y|| / \sigma)$**
- **Sigmoid kernel: $K(x, y) = \tanh(\alpha * x^T y + \beta)$**

Linear Kernel

In a regression problem, we are trying to estimate the optimal function to infer Y from X .

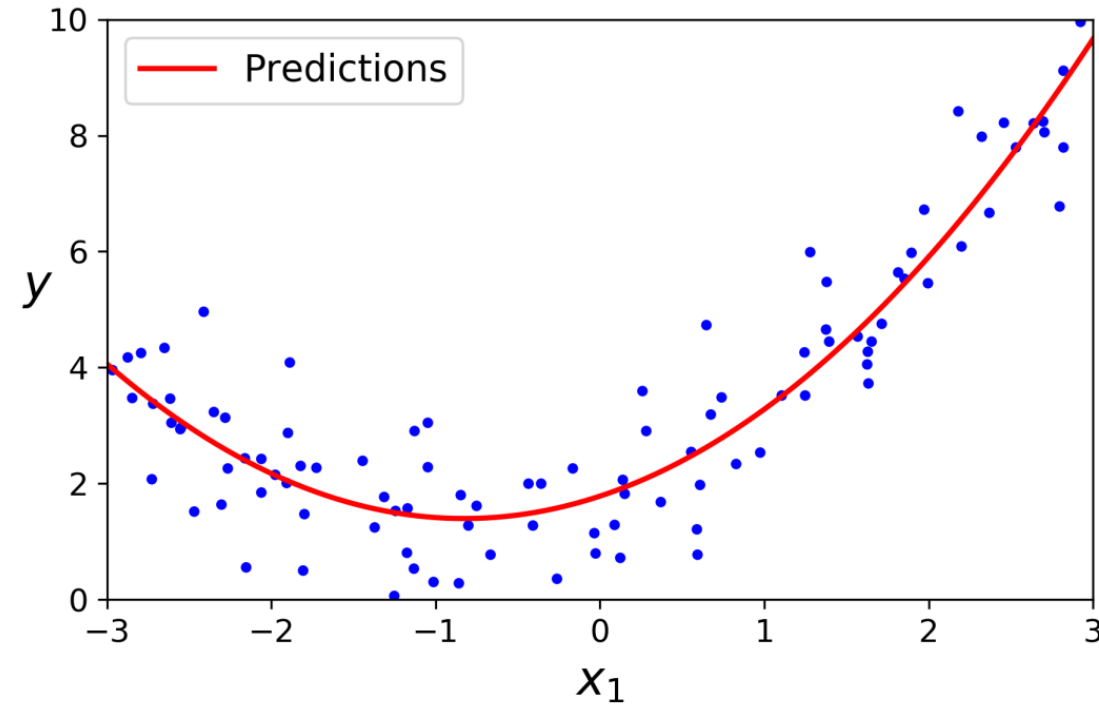
If we have a **non-linear relationship** between X and Y , we cannot simply fit a linear model on this data.

However, the goal of **kernel methods** is to use these non-linear relationship and still create a linear model.

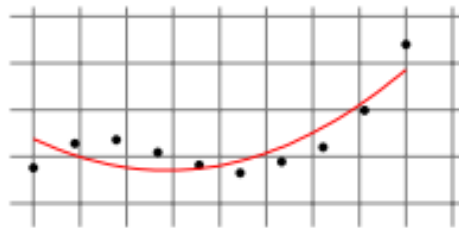
RBFs use Gaussian basis functions. Each basis function represents a Gaussian distribution in the input space.

Each data point is evaluated in all of the Gaussian distributions. The result is mapping the input vector from d to M dimensions.

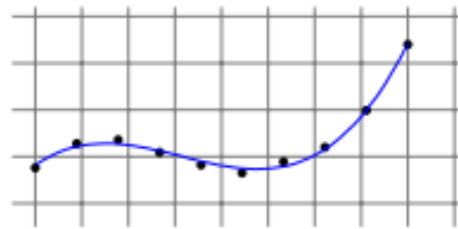
Polynomial Regression



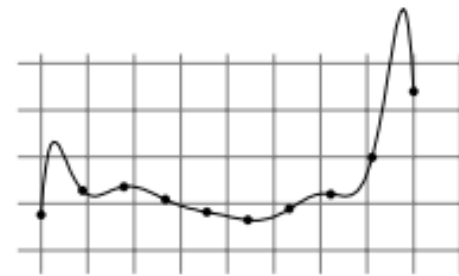
degree 2



degree 3

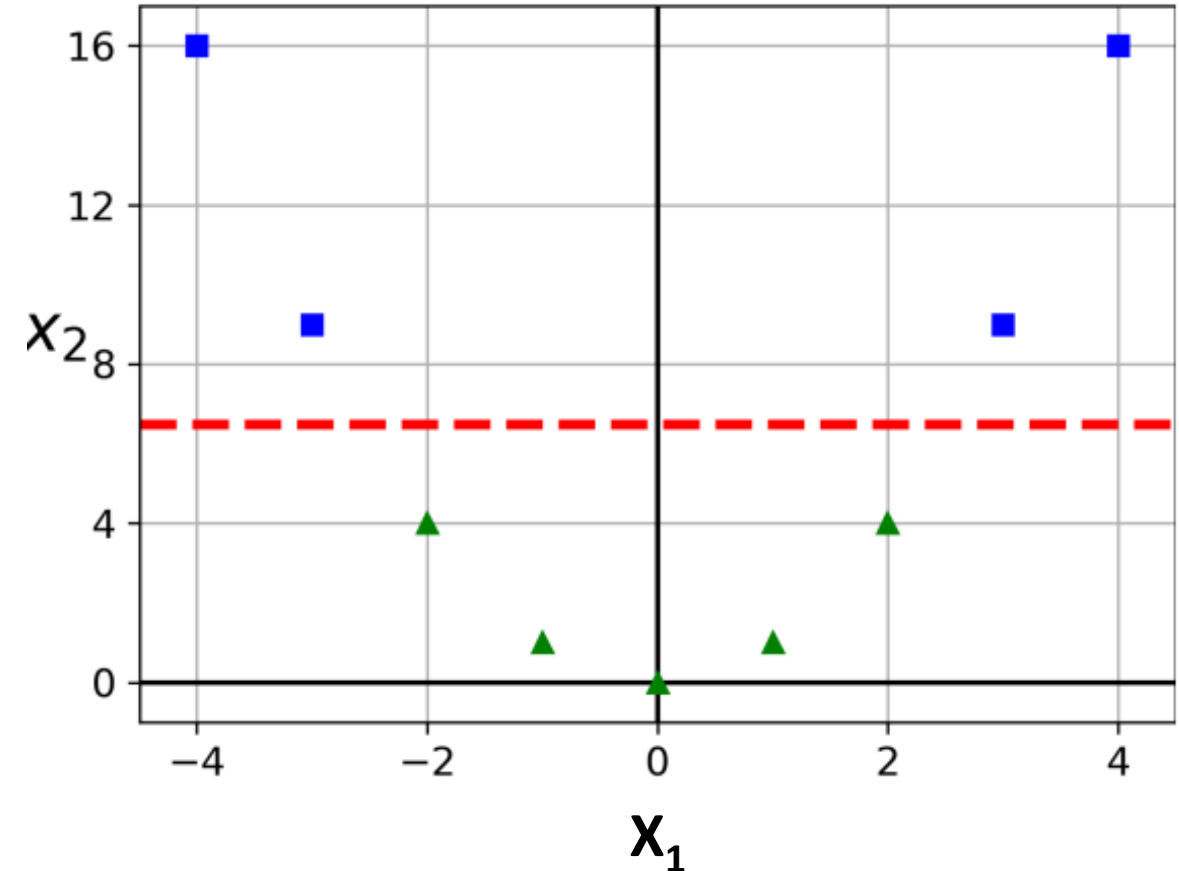
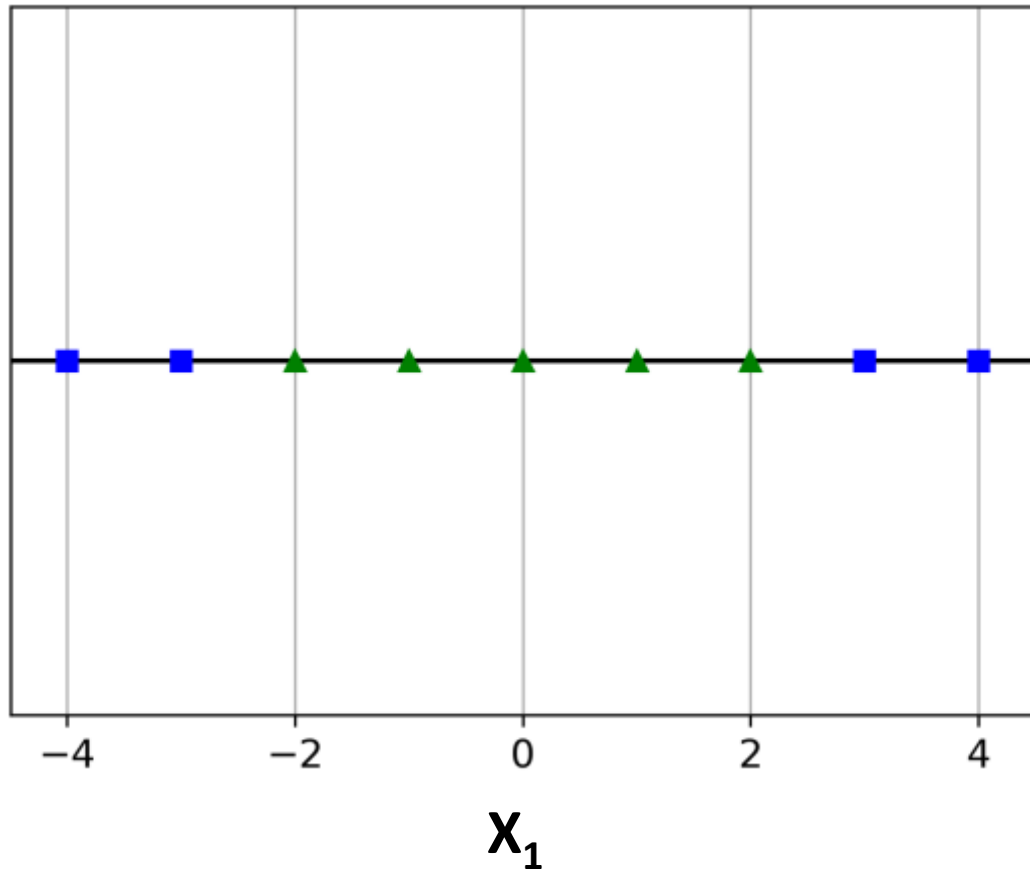


degree 10



Non-Linear to Linearly Separable Dataset

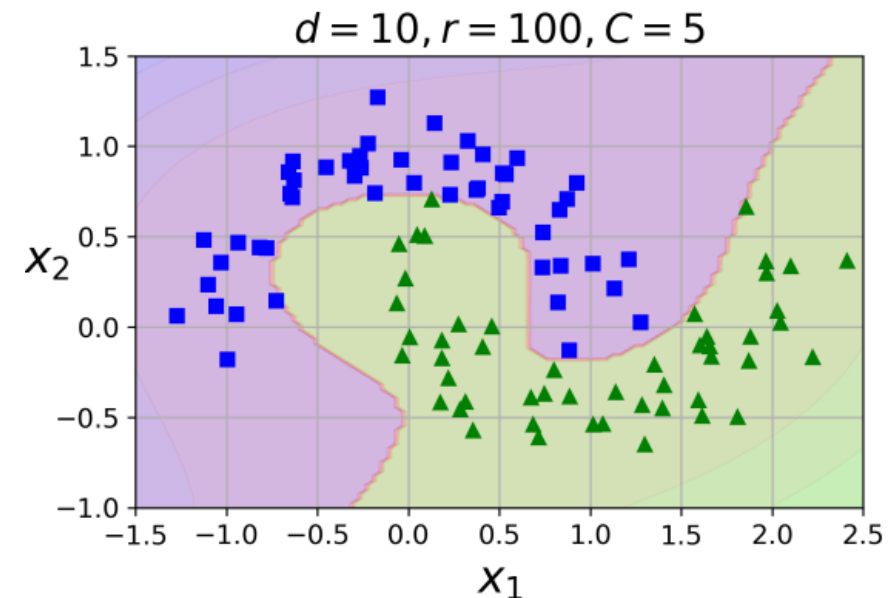
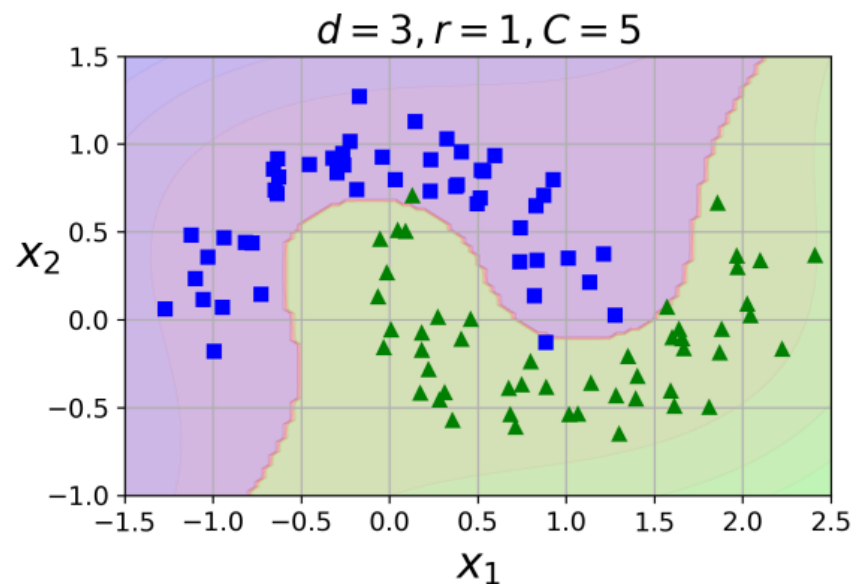
$$x_2 = (x_1)^2$$



SVM Classifier with Polynomial Kernel

A low polynomial degree it cannot deal with very complex datasets, and with a high polynomial degree it creates a huge number of features, making the model too slow.

```
from sklearn.svm import SVC
poly_kernel_svm_clf = Pipeline([
    ("scaler", StandardScaler()),
    ("svm_clf", SVC(kernel="poly", degree=3, coef0=1, C=5))
])
poly_kernel_svm_clf.fit(X, y)
```



Radial Basis Function

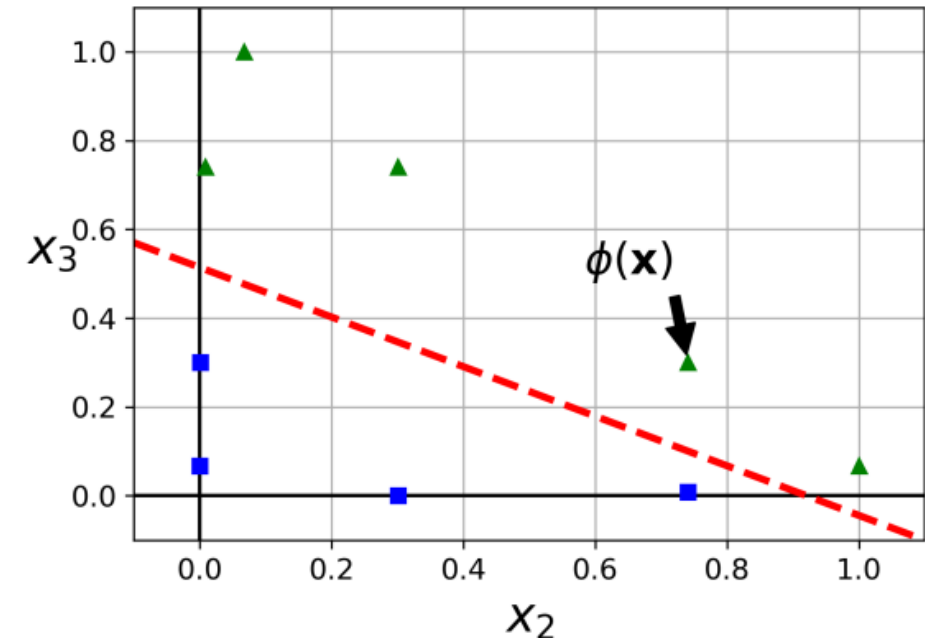
The polynomial features method may be computationally expensive to compute all the additional features for higher degrees, especially on large training sets

For larger datasets, Radial Basis Function (RBF) is more suited.

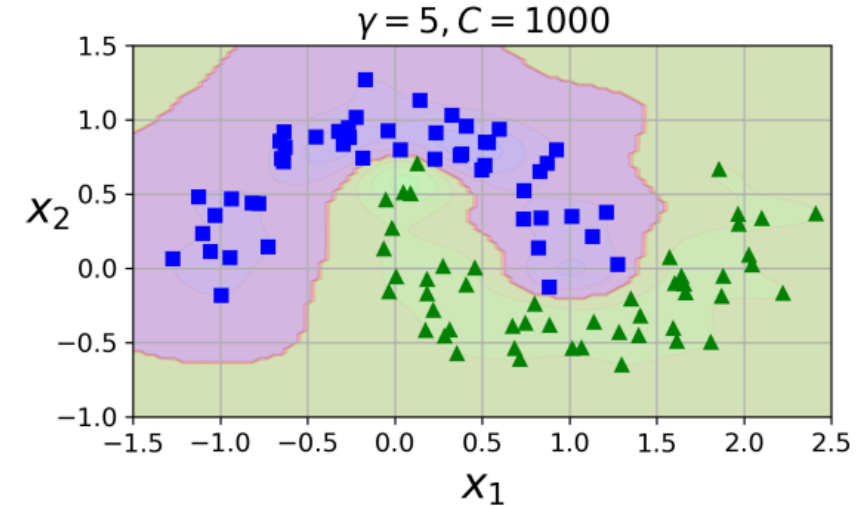
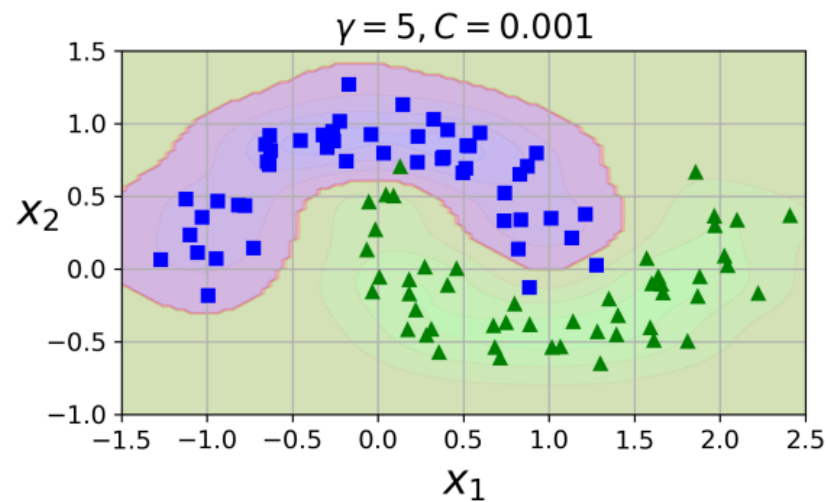
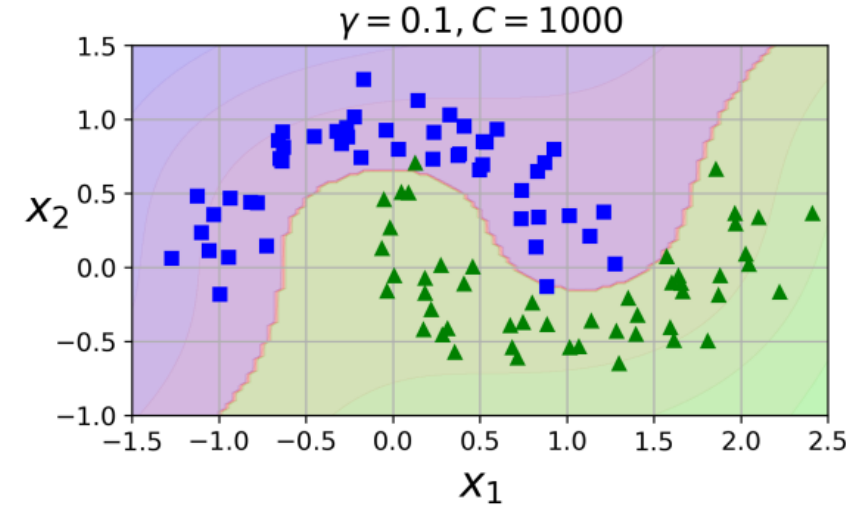
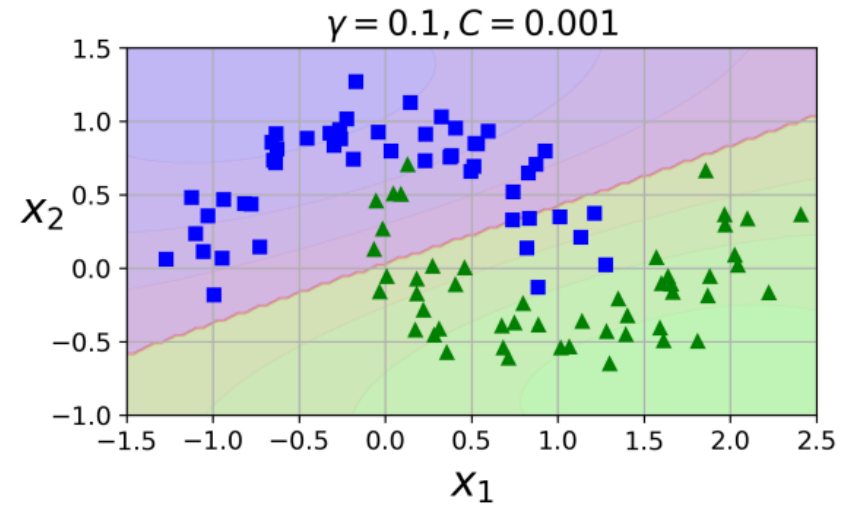
In RBF we have gamma ' γ ' as a hyperparameter.

γ acts like a regularization hyperparameter.

If the model is overfitting, we should reduce it, and if it is under- fitting, we should increase it (similar to the C hyperparameter)



SVM - Radial Basis Function

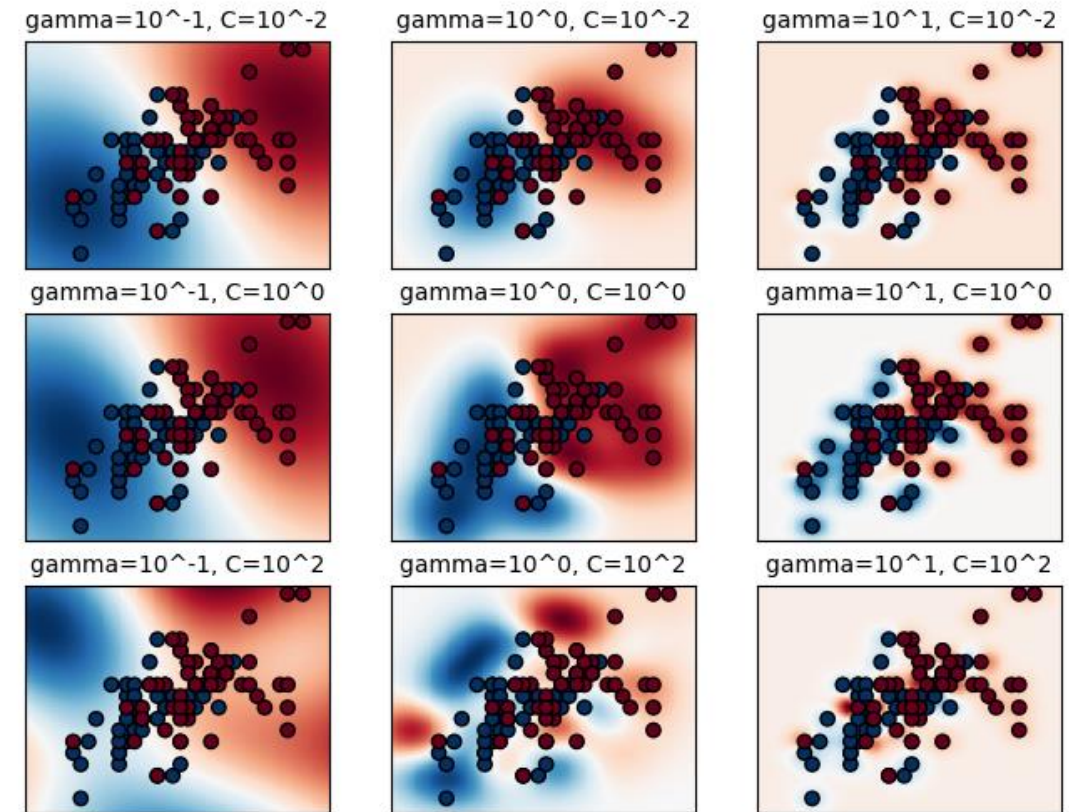


Radial Basis Function - γ Hyperparameter

The 'gamma' parameter defines how far the influence of a single training example reaches, with low values meaning 'far' and high values meaning 'close'.

The gamma parameters can be seen as the inverse of the radius of influence of samples selected by the model as support vectors.

γ : hyperparameter that regulates the non-linear kernel



Finding the right γ or σ along with the value of C is essential in order to achieve the best Bias-Variance Trade off.

Kernels Ridge Regression

Kernel Ridge Regression (KRR) is a kernel-based algorithm used for regression.

It is a **regularized** version of Ridge Regression that uses a kernel function to map the data into a higher-dimensional feature space.

KRR aims to find the **weights that minimize the sum of squared errors** between the predicted and actual values.

KRR is useful when the data is **not linearly separable**.

Kernels Methods - Selection

1. Understand the properties of different kernels:

- Different kernels have different properties and can be more suitable for certain types of data or tasks.
- Ex: The Gaussian kernel is commonly used in classification tasks, while the Laplacian kernel is often used in regression problems.

2. Choose a kernel that matches the data:

- The choice of kernel should be based on the underlying structure of the data.
- Ex: If the data is **linearly separable**, then the **linear kernel** may be a good choice.
- If the data is **not linearly separable**, then a **non-linear kernel**, such as the **Gaussian kernel**, may be more appropriate.

Kernels Methods - Selection

3. Adjust the kernel parameters:

- Most kernels have parameters that must be adjusted to achieve optimal performance.
- Ex: The Gaussian kernel has a bandwidth parameter that controls the width of the kernel. Choosing the right parameter values can have a significant impact on the performance of the model.

4. Consider computational efficiency:

- Some kernels can be computationally expensive, especially for large datasets.
- In such cases, it may be necessary to choose a simpler kernel to reduce the computational cost.

Kernels Methods - Selection

5. Understand the limitations of kernel methods:

- While kernel methods are very powerful and flexible, they do have some limitations.
- For instance, they may not perform well on high-dimensional data or on datasets with a large number of samples.

In summary, choosing the right kernel is an important part of designing a kernel method.

It requires a good understanding of the properties of different kernels, the structure of the data, and the computational requirements of the task.

Kernel Methods - Advantages and Disadvantages

Advantages:

- Can handle nonlinearity and high-dimensional data.
- Manage noisy and incomplete data.
- Generally have good generalization performance.

Disadvantages:

- Can be computationally expensive, especially for large datasets.
- Kernel functions can be difficult to choose and optimize.
- SVMs can be sensitive to the choice of hyperparameters.

Applications of Kernel Methods

- Image and speech recognition
- Text classification
- Bioinformatics
- Finance
- Astronomy
- Chemistry

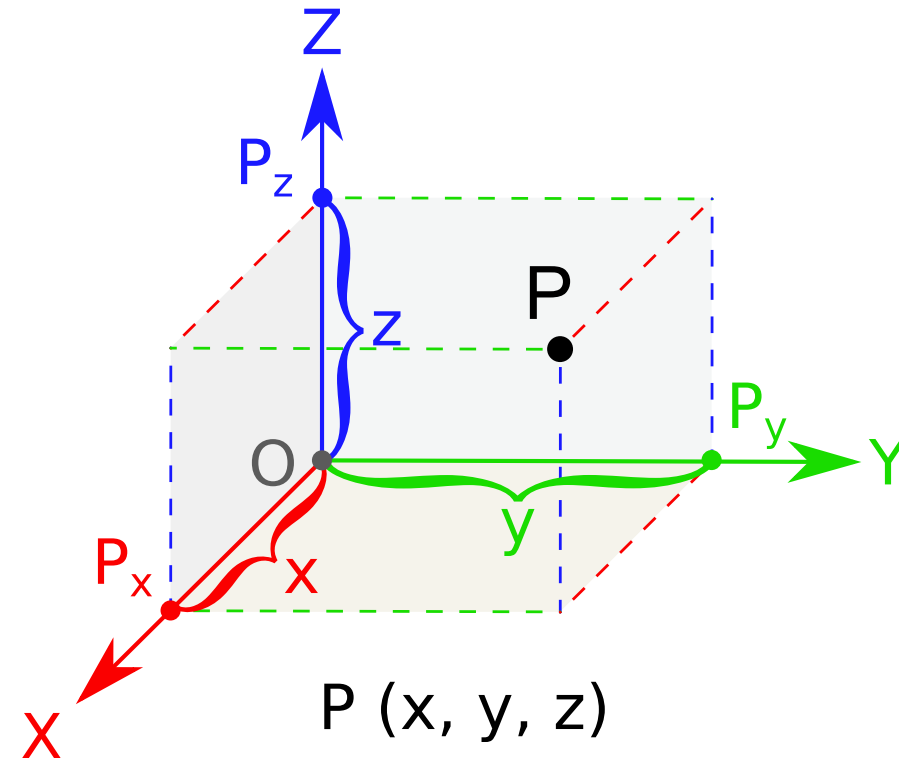
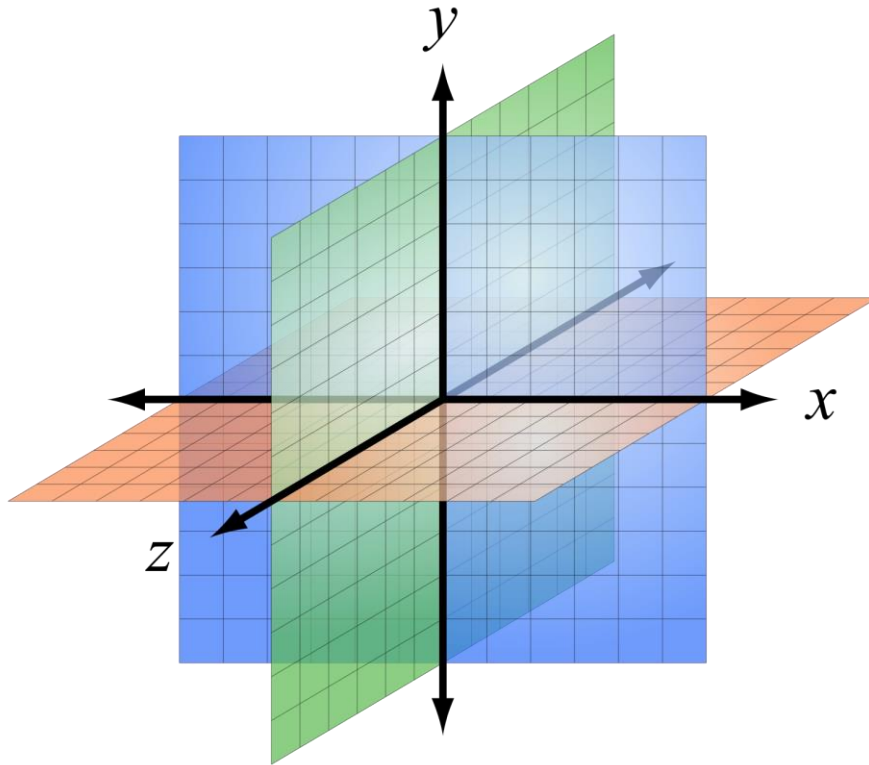
Summary

Kernel methods are powerful algorithms that can handle nonlinearity and high-dimensional data.

They are widely used in various applications and have good generalization performance.

However, they can be computationally expensive and require careful selection and optimization of kernel functions and hyperparameters.

Euclidean Space



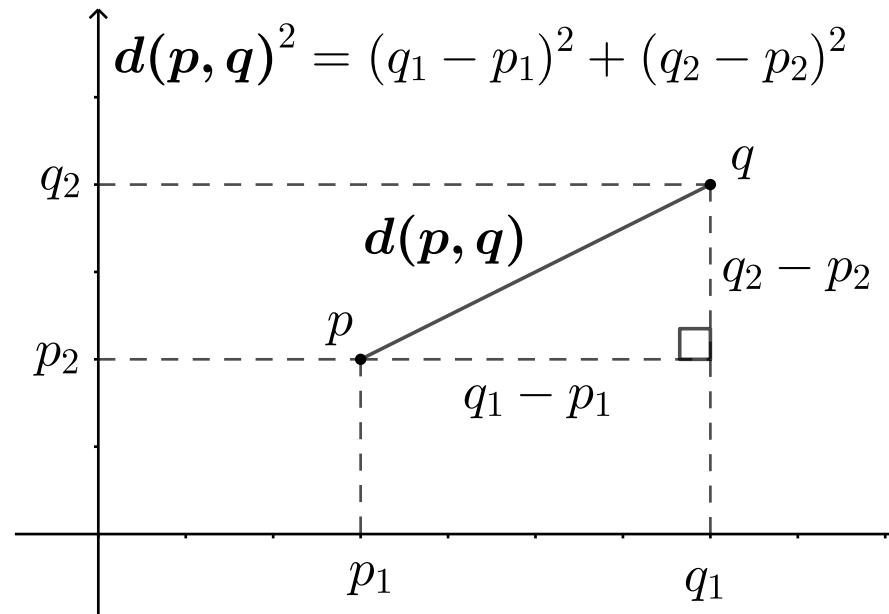
Euclidean space is the fundamental space of geometry, intended to represent physical space.

Euclidean spaces of any positive integer dimension n , are called Euclidean n -spaces when one wants to specify their dimension.

Euclidean Distance

In mathematics, the Euclidean distance between two points in Euclidean space is the length of a line segment between the two points.

Euclidean Distance can be calculated from the Cartesian coordinates of the points using the Pythagorean theorem



KNN

The k-nearest neighbors (KNN) algorithm is a non-parametric, supervised learning classifier that uses proximity to make classifications

It is one of the popular and simplest classification algorithms used in machine learning today.

A class label is assigned on the basis of a majority vote—i.e. the label that is most frequently represented around a given data point is used.

The distinction between these terminologies is that “majority voting” technically requires a majority of greater than 50%, which primarily works when there are only two categories.

When you have multiple classes—e.g. four categories, assign a class label with a vote of greater than 25%

KNN

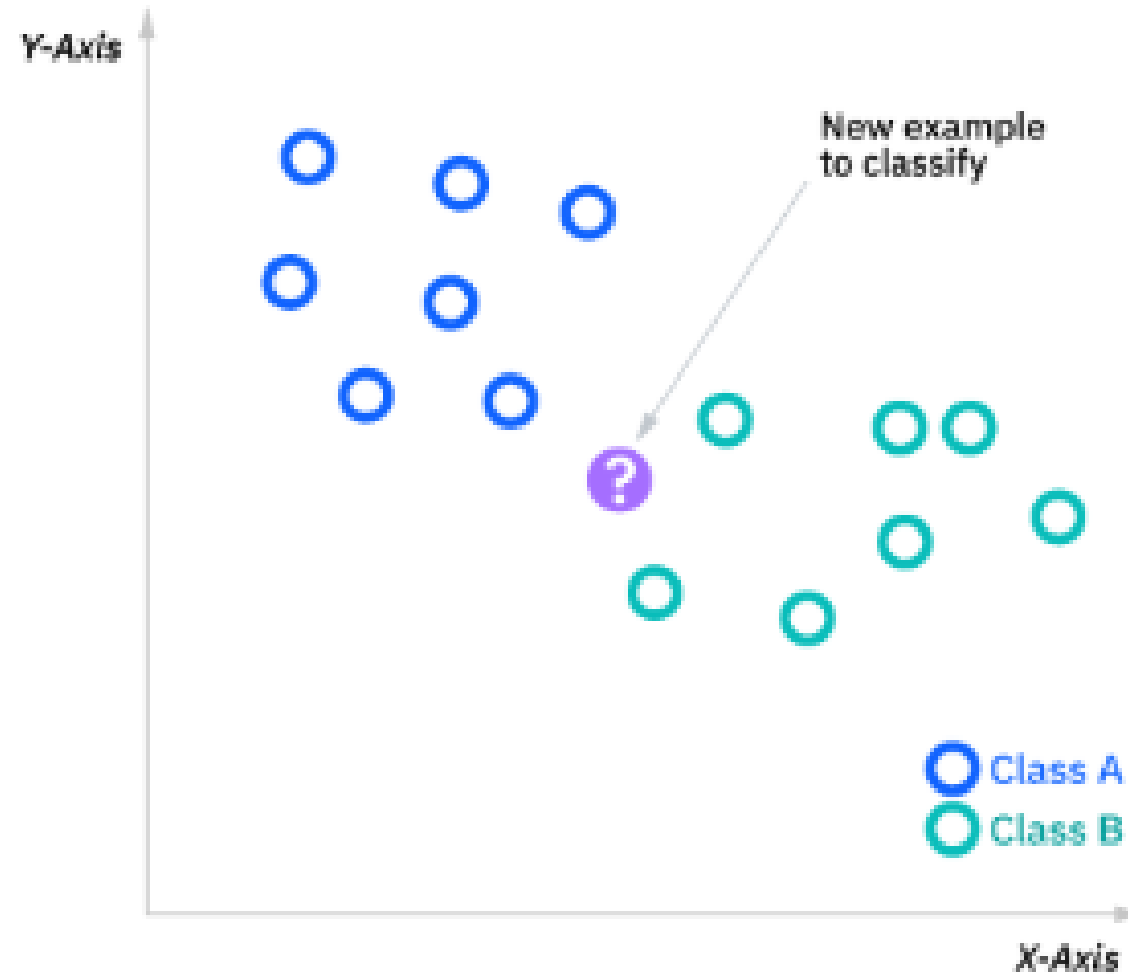
The KNN algorithm is also part of a family of “lazy learning” models, meaning that it only stores a training dataset versus undergoing a training stage.

This also means that all the computation occurs when a classification or prediction is being made.

Since it heavily relies on memory to store all its training data, it is also referred to as an instance-based or memory-based learning method.

Euclidean distance: This is the most commonly used distance measure, and it is limited to real-valued vectors.

KNN Classifying a Data Point



KNN Classifying a Data Point



KNN Classifying a Data Point



K-NN Advantages

The k value in the k -NN algorithm defines how many neighbors will be checked to determine the classification of a specific query point.

For example, if $k=1$, the instance will be assigned to the same class as its single nearest neighbor.

Defining k can be a balancing act as different values can lead to overfitting or underfitting.

Lower values of k can have high variance, but low bias, and larger values of k may lead to high bias and lower variance.

The choice of k will largely depend on the input data as data with more outliers or noise will likely perform better with higher values of k .

Overall, it is recommended to have an odd number for k to avoid ties in classification, and cross-validation tactics can help you choose the optimal k for your dataset.

KNN - Python

```
from sklearn.neighbors import KNeighborsClassifier
```

```
model_name = 'K-Nearest Neighbor Classifier'
```

```
knnClassifier = KNeighborsClassifier(n_neighbors = 5, metric = 'euclidean')
```

```
knn_model = Pipeline(steps=[('preprocessor', preprocessorForFeatures), ('classifier',  
knnClassifier)])
```

```
knn_model.fit(X_train, y_train)
```

```
y_pred = knn_model.predict(X_test)
```

K-NN Advantages

Like other machine learning algorithms, k-NN has its strengths and weaknesses.

Easy to implement:

Given the algorithm's simplicity and accuracy, it is one of the first classifiers that a new data scientist will learn.

Adapts easily:

As new training samples are added, the algorithm adjusts to account for any new data since all training data is stored into memory.

Few hyperparameters:

KNN only requires a k value and a distance metric, which is low when compared to other machine learning algorithms.

K-NN Disadvantages

Does not scale well:

Since KNN is a lazy algorithm, it takes up more memory and data storage compared to other classifiers. This can be costly from both a time and money perspective.

Curse of dimensionality:

The KNN algorithm tends to fall victim to the curse of dimensionality, which means that it doesn't perform well with high-dimensional data inputs.

Prone to overfitting:

Due to the “curse of dimensionality”, KNN is also more prone to overfitting. However, if the value of k is too high, it can underfit the data.

Applications of K-NN

Recommendation Engines:

The KNN algorithm has been used to provide automatic recommendations to users on additional content using clickstream data from websites.

Financial:

KNN on credit data can help banks assess loan risk to an organization or individual. It is used to determine the credit-worthiness of a loan applicant.

Healthcare:

KNN has also had applications within the healthcare industry, predicting the risk of heart attacks and prostate cancer.

Pattern Recognition:

KNN has also assisted in identifying patterns, such as in text and digit classification

END
